2. Synthetic Dataset (RAG Reference PDF)

For the agent to use the EOQ tool effectively, it needs to know your facility's internal costs. This synthetic document acts as the financial baseline for the RAG system.

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Section A: Standardized Cost Metrics
To standardize replenishment algorithms across Region 3, all automated EOQ calculations must utilize the following baseline financial metrics unless overridden by a specific vendor contract:

Metric	Standard Value	Description
Cost Per Order (S)	$45.00	Administrative cost to process, transmit, and receive a standard vendor PO.
Holding Cost (H) - Dry	$2.50	Annual cost to store one unit of dry grocery (accounts for space, utilities, and standard shrink).
Holding Cost (H) - Cold	$6.50	Annual cost to store one unit of refrigerated/frozen goods.

Section B: Replenishment Formulas & Policy

- Economic Order Quantity (EOQ): Used to determine the most cost-effective order size. Formula: `sqrt((2 * Annual Demand * Cost Per Order) / Holding Cost)`.
- Reorder Point (ROP): Calculates the exact inventory level that should trigger a new order. Formula: `(Daily Demand * Lead Time in Days) + Safety Stock`.
- Safety Stock Minimums: All critical holiday items must maintain a minimum safety stock of 15 units.